

Bugs found: 4

1. Off-by-one error in the for loop condition ($i \leq 5$) causes a buffer overflow when writing to `dest[5]`.
2. The string in `dest` is not null-terminated before being passed to `printf`.
3. Overwriting `dest[4]` with 'X' removes the possibility of a null terminator at that index, ensuring the string is improperly terminated.
4. Memory leak occurs because the dynamically allocated memory pointed to by `str` is never freed.